US012406742B2

(12) United States Patent
Lin et al.

(10) Patent No.: US 12,406,742 B2
(45) Date of Patent: Sep. 2, 2025

(54) **MEMORY DEVICE FOR PERFORMING IN-MEMORY COMPUTATION AND OPERATING METHOD THEREOF**

(71) Applicant: **MACRONIX INTERNATIONAL CO., LTD.**, Hsinchu (TW)

(72) Inventors: **Yu-Yu Lin**, New Taipei (TW); **Feng-Min Lee**, Hsinchu (TW)

(73) Assignee: **MACRONIX INTERNATIONAL CO., LTD.**, Hsinchu (TW)

( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 236 days.

(21) Appl. No.: **18/312,630**

(22) Filed: **May 5, 2023**

(65) **Prior Publication Data**

US 2024/0371453 A1 Nov. 7, 2024

(51) **Int. Cl.**

***G11C 17/18*** (2006.01)
***G06F 7/544*** (2006.01)
***G11C 17/16*** (2006.01)
*G11C 13/00* (2006.01)

(52) **U.S. Cl.**

CPC ............ ***G11C 17/18*** (2013.01); ***G06F 7/5443*** (2013.01); ***G11C 17/16*** (2013.01); *G11C 13/0021* (2013.01)

(58) **Field of Classification Search**

CPC ............ G11C 13/0064; G11C 13/0004; G11C 2013/0073; G11C 2013/0088; G11C 13/0007; G11C 13/0061; G11C 2013/0083; G11C 13/0069

USPC .......................................................... 365/96

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2019/0244662 A1* 8/2019 Lee ........................ H10B 63/30
2019/0286419 A1 9/2019 Lin
2022/0189526 A1 6/2022 Gong

FOREIGN PATENT DOCUMENTS

CN 111462798 A 7/2020
TW 1682388 B 1/2020
TW 202301336 A 1/2023

* cited by examiner

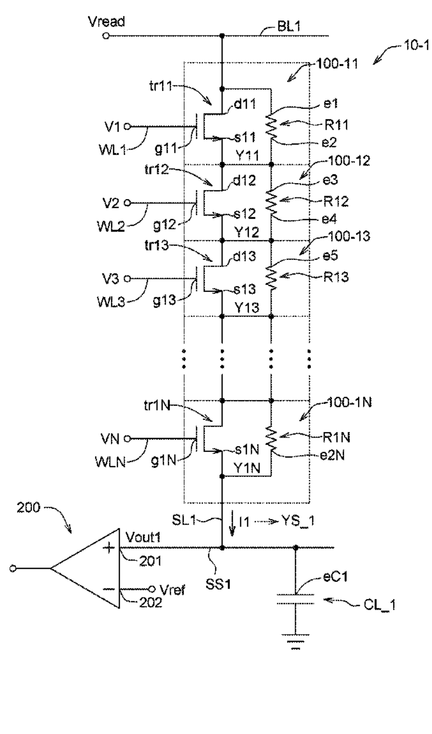
*Primary Examiner* — Amir Zarabian

(74) *Attorney, Agent, or Firm* — McClure, Qualey & Rodack, LLP

(57) **ABSTRACT**

A memory device for performing an in-memory computation, comprising a plurality of memory cells each stores a weight value and comprises a transistor and a resistor. A gate of the transistor receives an input voltage, the input voltage indicates an input value. When the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage. The resistor is connected to a drain and a source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value. Each of the memory cells performs a product computation of the input value and the weight value.

**16 Claims, 6 Drawing Sheets**



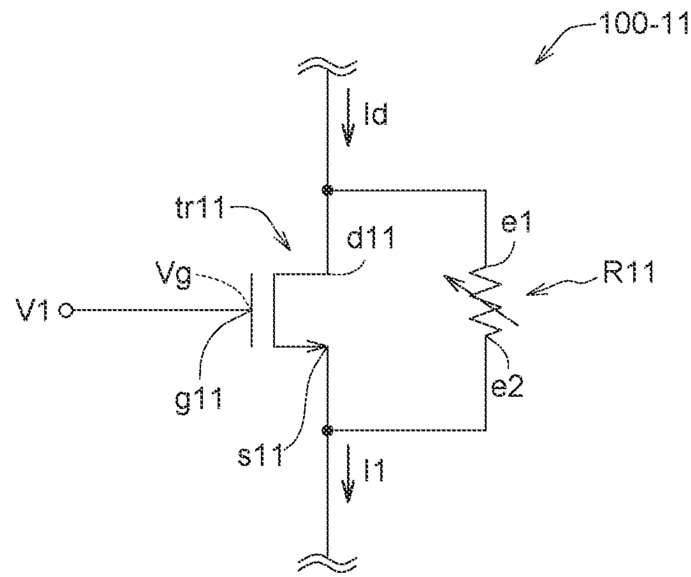



FIG. 1

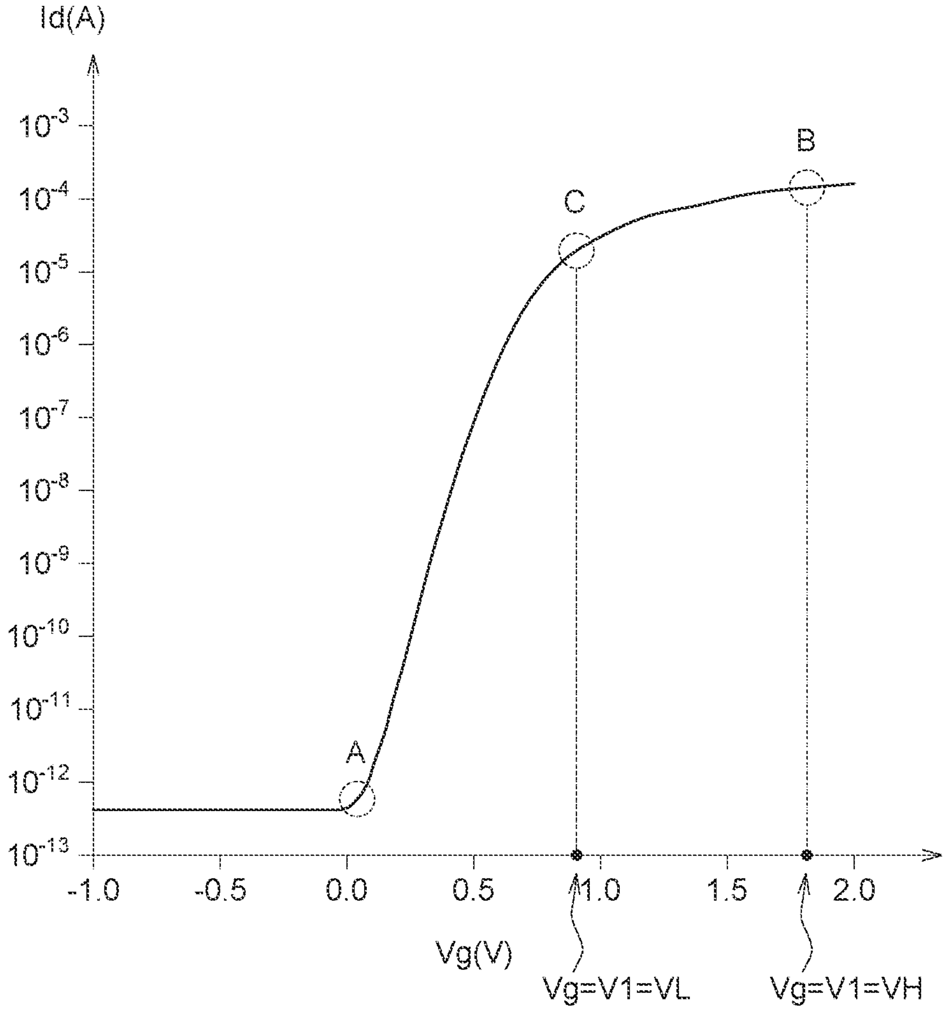

Id(A)
10^-3
10^-4
10^-5
10^-6
10^-7
10^-8
10^-9
10^-10
10^-11
10^-12
10^-13
A
B
C
-1.0
-0.5
0.0
0.5
1.0
1.5
2.0
Vg(V)
Vg=V1=VL
Vg=V1=VH

FIG. 2

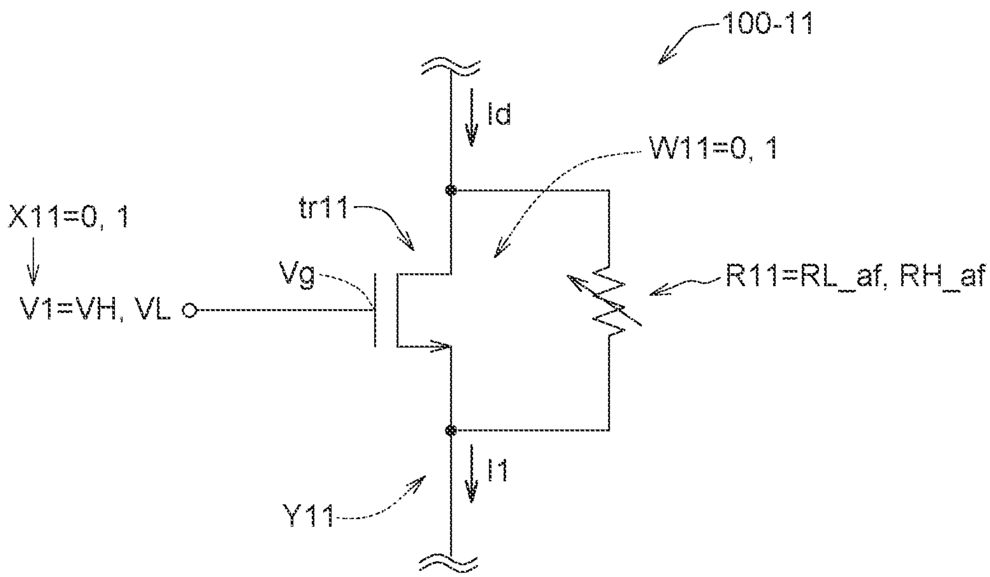
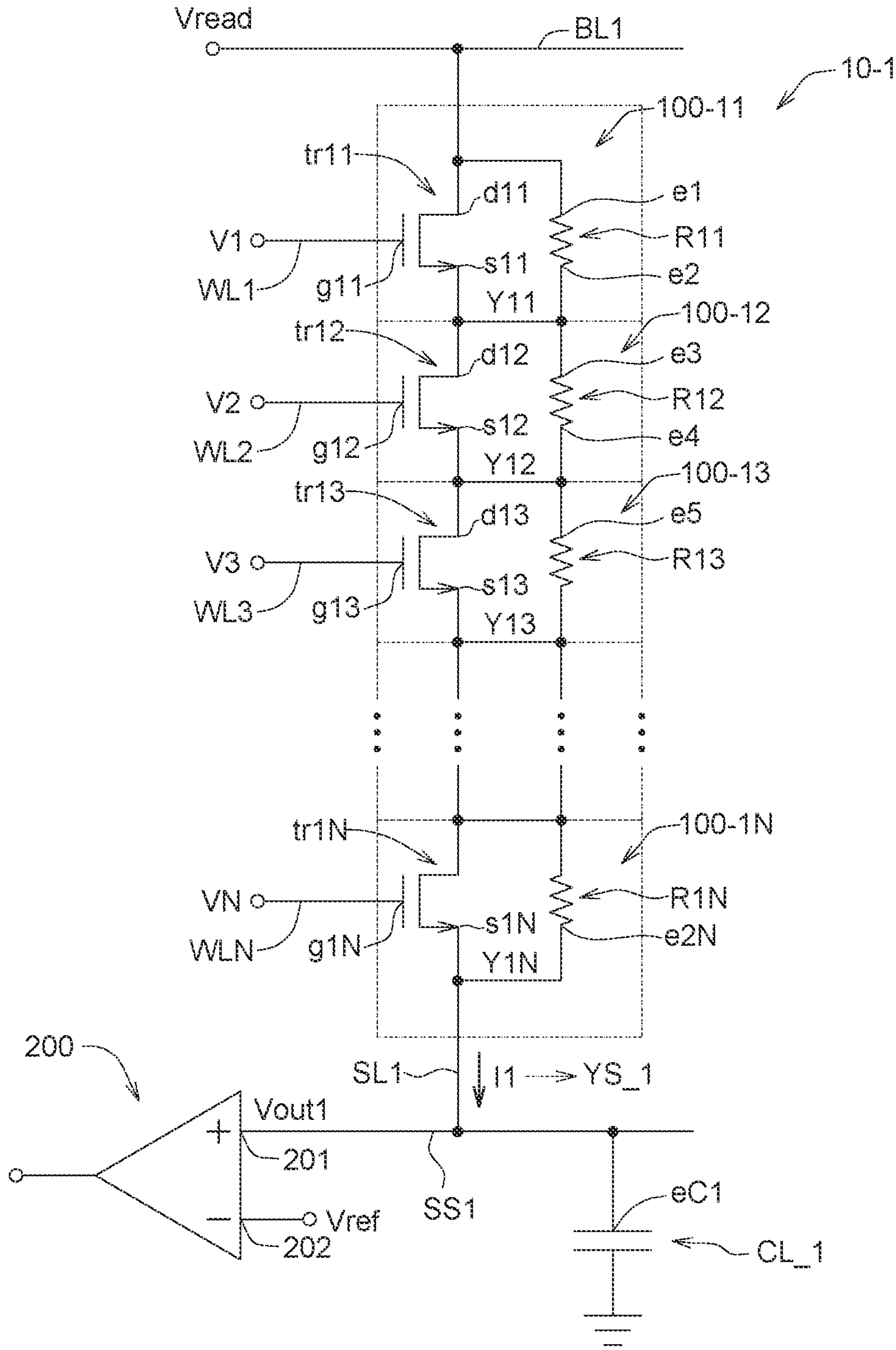



FIG. 3



Vread
BL1
10-1
100-11
tr11
d11
e1
V1
R11
WL1
g11
s11
e2
Y11
100-12
tr12
d12
e3
V2
R12
WL2
g12
s12
e4
Y12
100-13
tr13
d13
e5
V3
R13
WL3
g13
s13
Y13
tr1N
100-1N
VN
R1N
WLN
g1N
s1N
e2N
Y1N
200
SL1
I1
YS_1
Vout1
201
SS1
eC1
Vref
202
CL_1


FIG. 4

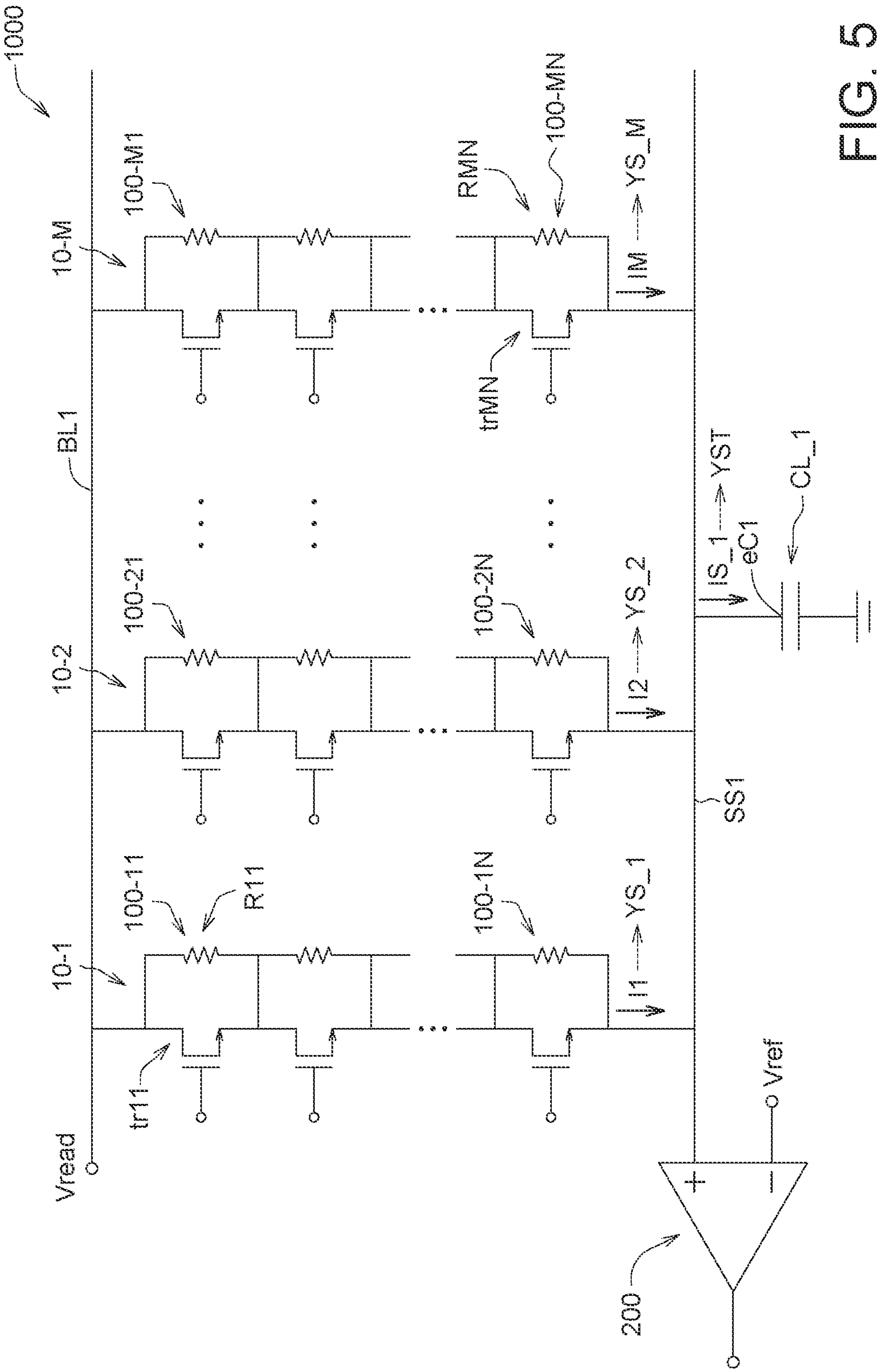

1000
10-1
10-2
10-M
BL1
100-11
R11
100-1N
100-21
100-2N
100-M1
RMN
100-MN
tr11
trMN
I1
I2
IM
YS_1
YS_2
YS_M
IS_1
YST
eC1
CL_1
SS1
Vread
Vref
200
+
−

FIG. 5

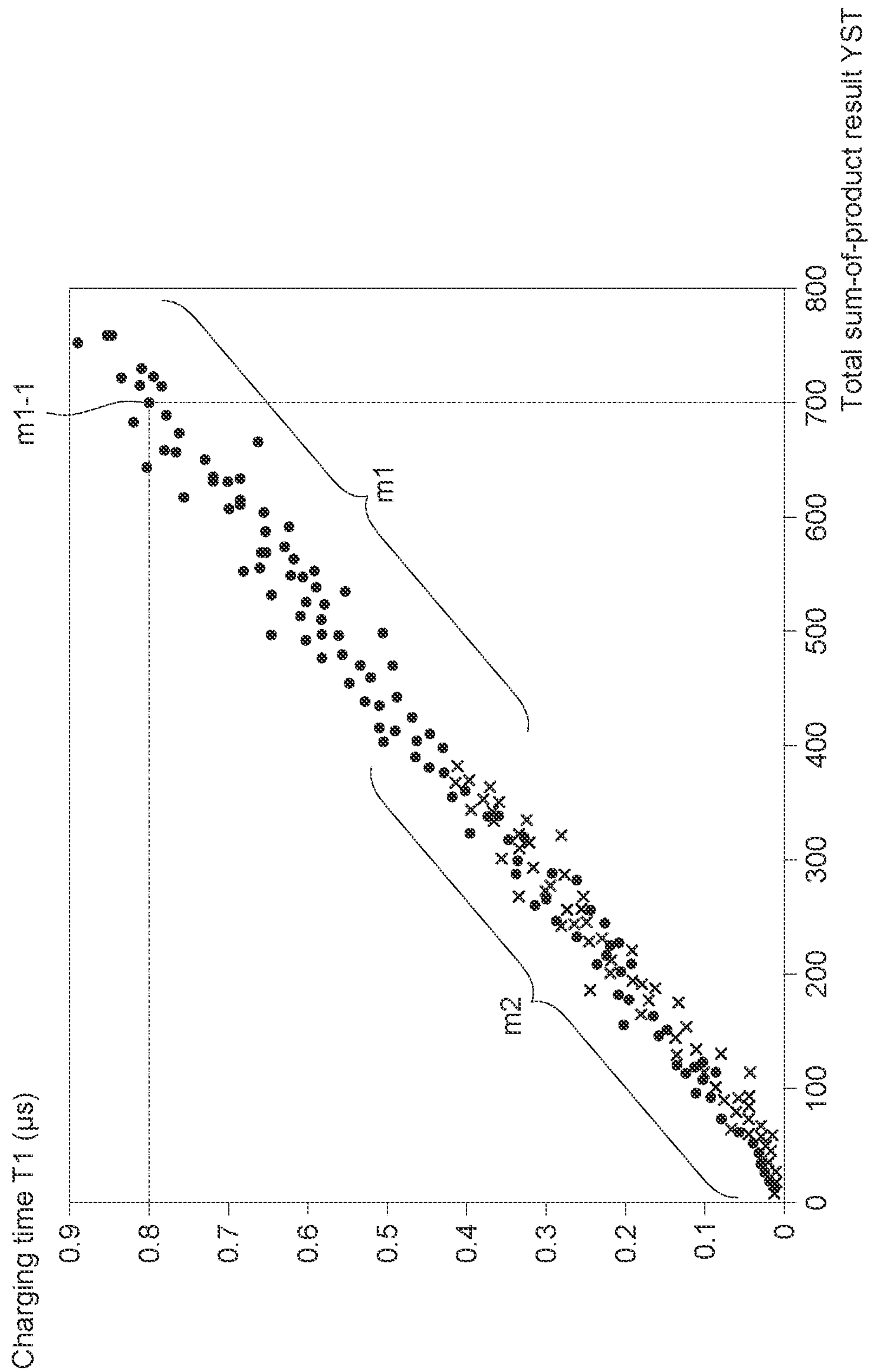

Charging time T1 (μs)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Total sum-of-product result YST
0
100
200
300
400
500
600
700
800
m1-1
m1
m2

FIG. 6

# MEMORY DEVICE FOR PERFORMING IN-MEMORY COMPUTATION AND OPERATING METHOD THEREOF

## TECHNICAL FIELD

The disclosure relates to a semiconductor device and an operating method thereof, and more particularly, relates to a memory device for performing in-memory computation and an operating method thereof.

## BACKGROUND

Artificial intelligence technology and big data processing technology have been widely used in all aspects of daily life. Artificial intelligence technology and big data processing technology rely on a large number of mathematical computations, such as neural network computations.

With the progress of semiconductor technology, in order to increase the computation speed, the above-mentioned mathematical computations may be performed inside the memory device, which is referred to as in-memory computation (IMC). In-memory computations include, for example, data searching, data comparison and sum-of-product computation.

In the sum-of-product computation, the memory cells of the memory device store weight values inside, and the memory cells receive input values from the outside. The memory cells perform sum-of-product computations based on the weight values and the input values, and the memory cells have a processing mechanism to change the weight values.

Those skilled in the art are devoted to improve the process and materials of the memory cells, expecting to capable of manufacturing the memory cells without additional masks, while the memory cells may still flexibly change the weight values.

## SUMMARY

According to an aspect of the disclosure, a memory device for performing an in-memory computation is provided. The memory device comprises a plurality of memory cells, each of the memory cells stores a weight value, and each of the memory cells comprises a transistor and a resistor. The transistor has a gate, a drain and a source, the gate receives an input voltage, the input voltage indicates an input value, when the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage, the second input voltage is higher than the first input voltage. The resistor is connected to the drain and the source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value. Each of the memory cells performs a product computation of the input value and the weight value.

According to another aspect of the disclosure, an operating method of a memory device for performing an in-memory computation is provided. The memory device comprises a plurality of memory cells, each of the memory cells comprises a transistor and a resistor, the transistor has a gate, a drain and a source, the resistor is connected to the drain and the source of the transistor, and the operating method comprises the following steps. Receiving an input voltage through the gate of the transistor, the input voltage indicates an input value. Setting the input voltage as a first input voltage, so as to operate the transistor at a first operating point. Setting the input voltage as a second input voltage, so as to operate the transistor at a second operating point, the second input voltage is higher than the first input voltage. Storing a weight value by each of the memory cells. Operating the resistor in a first state, so as to set the weight value as a first weight value. Operating the resistor in a second state, so as to set the weight value as a second weight value. Performing a product computation of the input value and the weight value by each of the memory cells.



## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic diagram of a memory cell according to an embodiment of the disclosure.

FIG. **2** is a current-voltage curve diagram of the drain current and the gate voltage of the transistor in FIG. **1**.

FIG. **3** is a schematic diagram of the memory cell in FIG. **1** performing computations.

FIG. **4** is a schematic diagram of the memory string performing computations, according to an embodiment of the disclosure.

FIG. **5** is a schematic diagram of the memory device performing computations, according to an embodiment of the disclosure.

FIG. **6** is a measurement diagram of the computation results of the memory device in FIG. **5**.

In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically illustrated in order to simplify the drawing.

## DETAILED DESCRIPTION

FIG. **1** is a schematic diagram of a memory cell **100-11** according to an embodiment of the disclosure. The memory cell **100-11** includes a transistor tr**11** and a resistor R**11**, the transistor tr**11** is connected to the resistor R**11** in parallel. The transistor tr**11** has a gate g**11**, a source s**11** and a drain d**11**. The drain d**11** is connected to the first end e**1** of the resistor R**11**, and the source s**11** is connected to the second end e**2** of the resistor R**11**. The gate g**11** receives the input voltage V**1**, and the gate voltage Vg of the gate g**11** is equal to the input voltage V**1**.

The drain d**11** of the transistor tr**11** has a drain current Id, and the memory cell **100-11** has an output current I**1**. The output current I**1** is equal to the sum of the drain current Id and the current of the resistor R**11**.

In one example, a plurality of memory cells may form a memory string. For example, the memory cell **100-11** and other memory cells (FIG. **1** does not show other memory cells) are connected in series to form a memory string. The source s**11** of the transistor tr**11** of the memory cell **100-11** may be coupled to the drain of the transistor of a neighboring memory cell. The drain current of the transistor of and the current of the resistor of each of the memory cells in the memory string are summed up to form the output current I**1** of the memory string. The computation result of the sum-of-product computation performed by the memory string may be calculated according to the output current I**1** (which will be described in detail in embodiments of following paragraphs with FIG. **4**).

The resistor **R11** is formed of a material having resistive characteristics. In one example, the resistor **R11** is, for example, an "anti-fuse". In other examples, the resistor **R11** is, for example, a one-time programmable memory (OTP memory), a multiple-time programmable memory (MTP memory), a Resistive Random Access Memory (ReRAM), or other types of non-volatile memory. Taking the anti-fuse as an example, the resistor **R11** implements the characteristics of anti-fuse through the gate oxide breakdown mechanism. The resistor **R11** operates in an initial state S_I and a programmed state S_P. The initial state S_I may be referred to as a "first state", and the programmed state S_P may be referred to as a "second state".

In the initial state S_I, the resistor **R11** is equivalent to a state of being fused, and the resistor **R11** is nearly being in a state of turned-OFF. Therefore, the resistor **R11** has a first resistance value RH_af which is relatively high. On the other hand, after the programming operation is applied to the resistor **R11**, the gate oxide layer of the resistor **R11** breaks down in response to the applied programming voltage, the state of the resistor **R11** is converted to the programmed state S_P, and the resistor **R11** is nearly being in a state of turned-ON. Therefore, the resistor **R11** has a second resistance value RL_af which is relatively low. The second resistance value RL_af is lower than the first resistance value RH_af, which conforms to the relationship shown in formula (1):

$$\mathrm{RH\_af} > \mathrm{RL\_af} \tag{1}$$

As shown in Table 1, when the state of the resistor **R11** is the initial state S_I, the resistor **R11** is in a high-resistance state (high-R state) and has a relatively high first resistance value RH_af. In contrast, when the state of the resistor **R11** is the programmed state S_P, the resistor **R11** is in a low-resistance state (low-R state) and has a relatively low second resistance value RL_af.

TABLE 1

| | | |
|---|---|---|
| State of resistor R11 | high-R state Initial state S_I | low-R state Programmed state S_P |
| Resistance value resistor R11 | The first resistance value RH_af | The second resistance value RL_af |

In another example, the resistor **R11** may also be realized by a "fuse". The characteristics of the fuse are opposite to those of the anti-fuse described above. When the resistor **R11** is a fuse, the resistor **R11** in the initial state S_I has a relatively low second resistance value RL_af. After the programming operation is performed to the resistor **R11**, the resistor **R11** is converted to the programmed state S_P, and the resistor **R11** has a relatively high first resistance value RH_af.

On the other hand, the transistor tr**11** is, for example, an N-type metal oxide semiconductor field effect transistor (N-type MOSFET), which is referred to as "N-type transistor". The gate g**11** of the transistor tr**11** receives the input voltage V**1**. When the input voltage V**1** is a relatively high voltage, the channel of the transistor tr**11** is substantially turned-on, and the transistor tr**11** has a relatively low equivalent resistance. In contrast, when the input voltage V**1** is a relatively low voltage, the channel of the transistor tr**11** is substantially turned-off, and the transistor tr**11** has a relatively high equivalent resistance value. The relationship between the input voltage V**1** and the equivalent resistance value of the transistor tr**11** will be described in detail below with reference to FIG. **2**.

FIG. **2** is a current-voltage curve diagram of drain current Id and gate voltage Vg of the transistor tr**11** in FIG. **1**. The gate voltage Vg of the transistor tr**11** is equal to the input voltage V**1**. On the current-voltage curve of the drain current Id and the gate voltage Vg, different operating points of the transistor tr**11** may be defined. At the operating point A, the input voltage V**1** received by the transistor tr**11** is 0V (i.e., the gate voltage Vg of the transistor tr**11** is 0V). Since the gate voltage Vg of the transistor tr**11** is very low, the channel of the transistor tr**11** is almost completely-turned-off, and the equivalent resistance value of the transistor tr**11** is very high. Therefore, the drain current Id of the transistor tr**11** is very low, and the drain current Id is about $10^{-12}$ A.

When the memory cell **100-11** and other memory cells are connected in series to form a memory string, the source of the transistor and the resistor of the last memory cell in the memory string may be connected to a load capacitor CL_**1** (not shown in FIG. **1**). The drain current Id of the transistor of each of the memory cells in the memory string is added to the current of the resistor to form the output current I**1** of the memory string, and the output current I**1** charges the load capacitor CL_**1**.

The drain current Id at the operating point A has a much less current amount (approximately $5\times10^{-13}$ A), hence the current amount of the output current I**1** of the memory string is much less, and the charging time T**1** of the load capacitor CL_**1** will be very long and exceed a reasonable range for measurement. Therefore, the transistor tr**11** of this embodiment is not operated at the operating point A, instead, the input voltage V**1** is increased (i.e., the gate voltage Vg is increased), so that the transistor tr**11** is slightly turned-on, causing the transistor tr**11** to operate at operating point C. The operating point C may be referred to as a "first operating point". Since the transistor tr**11** is slightly turned-on, the transistor tr**11** has a first equivalent resistance value RH_tr which is relatively high.

When the transistor tr**11** is an N-type transistor, the gate voltage Vg at the operating point C is higher than the threshold voltage Vt of the transistor tr**11** multiplied by a first ratio N1. In this embodiment, the first ratio N1 may be "½" or "⅓", and the gate voltage Vg at the operating point C is 0.9V, as shown in formula (2) or formula (3):

$$V1 = Vg = 0.9V > Vt \times 1/2 \tag{2}$$

$$V1 = Vg = 0.9V > Vt \times 1/3 \tag{3}$$

At the operating point C, in response to the gate voltage Vg of 0.9V, the current amount of the drain current Id increases to about $5\times10^{-5}$ A, hence the output current I**1** of the memory string has an increased current amount, and the charging time T**1** for the drain current Id to charge the load capacitor CL_**1** may be within a reasonable range for measurement.

At the operating point C, the input voltage V**1** of the transistor tr**11** is equal to the first input voltage VL, and the first input voltage VL is 0.9V. The first input voltage VL is higher than the threshold voltage Vt multiplied by the first ratio N1.

On the other hand, the gate voltage Vg of the transistor tr**11** is further increased to 1.8V, such that the transistor tr**11** operates at the operating point B. The operating point B may be referred to as "the second operating point". At the operating point B, a greater degree of being-turned-on for the transistor tr**11** is achieved, and the equivalent resistance value of the transistor tr**11** is reduced to a second equivalent resistance value RL_tr. The second equivalent resistance value RL_tr is lower than the first equivalent resistance value RH_tr at the operating point C, which conforms to the relationship shown in formula (4-1):

$$RH\_tr > RL\_tr \tag{4-1}$$

At the operating point B, the drain current Id of the transistor tr**11** has a current amount increasing up to $5\times10^{-4}$ A. Therefore, the charging time T**1** for the output current I**1** of the memory string to charge the load capacitor CL_**1** is also within a reasonable range for measurement. The input voltage V**1** (equal to the gate voltage Vg) of the transistor tr**11** is raised to a second input voltage VH. The second input voltage VH is, for example, 1.8V, and the second input voltage VH is higher than the first input voltage VL.

In one example, the relationship of the second equivalent resistance value RL_tr of the transistor tr**11** at the operating point B and the first equivalent resistance value RH_tr at the operating point C may be defined according to a second ratio N2. For example, the first equivalent resistance value RH_tr is lower than or equal to a product of the second equivalent resistance value RL_tr and the second ration N2, wherein the second ration N2 is e.g., "1000", as shown in formula (4-2):

$$RH\_tr \leq RL\_tr \times N2 \tag{4-2}$$

According to formula (4-1) and formula (4-2), a relationship shown in formula (4-3) may be obtained:

$$RL\_tr < RH\_tr \leq RL\_tr > N2 \tag{4-3}$$

To sum up, the transistor tr**11** has different equivalent resistance values at different operating points B and C on the relationship curve of the drain current Id and gate voltage Vg of FIG. **2**.

At operating point C, the gate voltage Vg of the transistor tr**11** is 0.9V, and the drain current Id of the transistor tr**11** is approximately $5\times10^{-5}$ A, and the transistor tr**11** is in a state of high resistance and has a higher first equivalent resistance value RH_tr. On the other hand, at operating point B, the gate voltage Vg of the transistor tr**11** is 1.8V, and the drain current Id of the transistor tr**11** is approximately $5\times10^{-4}$ A, and the transistor tr**11** is in a state of low resistance and has a lower second equivalent resistance value RL_tr.

As shown in Table 2, at the operating point C, the input voltage V**1** is equal to the relatively lower first input voltage VL (i.e., the first input voltage VL is equal to the gate voltage Vg of 0.9V), the transistor tr**11** is in a high-resistance state having a first equivalent resistance RH_tr. In contrast, at the operating point B, the input voltage V**1** is equal to the relatively higher second input voltage VH (i.e., the second input voltage VH is equal to the gate voltage Vg of 1.8V), the transistor tr**11** is in a low-resistance state having a second equivalent resistance RL_tr.

TABLE 2

| State of transistor tr11 | high-R state | low-R state |
|---|---|---|
| Operating point of transistor tr11 | Operating point C | Operating point B |
| Input voltage V1 | The first input voltage VL (equal to the gate voltage Vg of 0.9 V) | The second input voltage VH (equal to the gate voltage Vg of 1.8 V) |
| Drain current Id of transistor tr11 | $5 \times 10^{-5}$ A | $5 \times 10^{-4}$ A |
| Equivalent resistance value of transistor tr11 | The first equivalent resistance value RH__tr | The second equivalent resistance value RL__tr |

The first resistance value RH_af and the second resistance value RL_af of the resistor R**11** are compared with the first equivalent resistance value RH_tr and the second equivalent resistance value RL_tr of the transistor tr**11**, the relationship shown in formula (5) may be obtained:

$$RH\_af > RH\_tr > RL\_tr > RL\_af \tag{5}$$

In another example, the second resistance value RL_af of the resistor R**11** is lower than the second equivalent resistance value RL_tr of the transistor tr**11**, and the relationship shown in formula (6) is obtained:

$$RRH\_af > RH\_tr > RL\_af > RL\_tr \tag{6}$$

FIG. **3** is a schematic diagram of the memory cell **100-11** in FIG. **1** performing computation. The memory cell **100-11** may perform a product computation (or multiplication computation). Referring to FIG. **3** (and refer to FIG. **2** in conjunction), the input voltage V**1** of the transistor tr**11** may represent the input value X**11** of the memory cell **100-11**. At the operating point C, the input voltage V**1** of the transistor tr**11** is equal to the first input voltage VL, which represents that the input value X**11** of the memory cell **100-11** is a bit of "1". At the operating point B, the input voltage V**1** of the transistor tr**11** is equal to the second input voltage VH, which represents that the input value X**11** of the memory cell **100-11** is a bit of "0".

On the other hand, the memory cell **100-11** stores a weight value W**11**, and the weight value W**11** is used to perform an in-memory computation (IMC) in the memory cell **100-11**. In this embodiment, in-memory computation is a product computation of the input value X**11** and the weight value W**11**. The weight value W**11** may be changed by changing the resistance value of the resistor R**11**, and the weight value W**11** may be equal to a "first weight value" or a "second weight value". In this embodiment, the first weight value is a bit of "1", and the second weight value is a bit of "0". The resistor R**11** has a first resistance value RH_af in the initial state S_I, at this time, the weight value W**11** is equal to the first weight value (i.e., "1"). On the other hand, the resistor R**11** has the second resistance value RL_af in the programmed state S_P, at this time, the weight value W**11** is equal to the second weight value (i.e., "0").

The value of the weight value (i.e., the first weight value or the second weight value) may be changed by changing the state of the resistor R**11** (i.e., the initial state S_I or the programmed state S_P) so that the resistor R**11** has different resistance values. Therefore, there is no need to dispose additional masks in the manufacturing process of the memory cell **100-11** to change the value of the weight value W**11**. The memory cell **100-11** may adopt manufacturing process of logic circuit, such that the memory cell **100-11** may realize the in-memory computation.

Table 3 shows four conditions of the product computation of the input value X**11** of the memory cell **100-11** and the weight value W**11**. In the first condition (cond-1), the input voltage V**1** of the transistor tr**11** is equal to the second input voltage VH, and the resistor R**11** is in the programmed state S_P, indicating that the input value X**11** is "0" and the weight value W**11** is "0". The transistor tr**11** has a second equivalent resistance value RL_tr, and the resistor R**11** has a second resistance value RL_af, and the total equivalent resistance value of the memory cell **100-11** is equal to a parallel computation of the second equivalent resistance value RL_tr and the second resistance value RL_af. According to formula (5), the second equivalent resistance value RL_tr is higher than the second resistance value RL_af, hence, the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second resistance value RL_af.

In the second condition (cond-2), the input voltage V**1** of the transistor tr**11** is equal to the first input voltage VL, and the resistor R**11** is in the programmed state S_P, indicating that the input value X**11** is "1" and the weight value W**11** is "0". The transistor tr**11** has a first equivalent resistance value RH_tr, and the resistor R**11** has a second resistance value RL_af, the total equivalent resistance value of the memory cell **100-11** is equal to the parallel computation of the first equivalent resistance value RH_tr and the second resistance value RL_af. According to formula (5), the first equivalent resistance value RH_tr is higher than the second resistance value RL_af, hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second resistance value RL_af.

In the third condition (cond-3), the input voltage V**1** of the transistor tr**11** is equal to the second input voltage VH, and the resistance R**11** is in the initial state S_I, indicating that the input value X**11** is "0" and the weight value W**11** is "1". The transistor tr**11** has a second equivalent resistance value RL_tr, and the resistor R**11** has a first resistance value RH_af, the total equivalent resistance value of the memory cell **100-11** is equal to a parallel computation of the second equivalent resistance value RL_tr and the first resistance value RH_af. According to formula (5), the second equivalent resistance value RL_tr is lower than the first resistance value RH_af, hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the second equivalent resistance value RL_tr.

In the fourth condition (cond-4), the input voltage V**1** of the transistor tr**11** is equal to the first input voltage VL, and the resistance R**11** is in the initial state S_I, indicating that the input value X**11** is "1" and the weight value W**11** is "1". The transistor tr**11** has a first equivalent resistance value RH_tr, and the resistor R**11** has a first resistance value RH_af, the total equivalent resistance value of the memory cell **100-11** is equal to a parallel computation of the first equivalent resistance value RH_tr and the first resistance value RH_af According to formula (5), the first equivalent resistance value RH_tr is lower than the first resistance value RH_af, hence the total equivalent resistance value of the memory cell **100-11** is substantially equal to the first equivalent resistance value RH_tr.

TABLE 3

| Product computation of the memory cell 100-1 | State of resistor R11<br>Programmed state S_P<br>Weight value W11 = "0"<br>(equal to the second weight value) | State of resistor R11<br>Initial state S_I<br>Weight value W11 = "1"<br>(equal to the first weight value) |
|---|---|---|
| Input voltage V1 = VH<br>Input value X11 = "0" | (cond-1)<br>RL_af//RL_tr ≅ RL_af<br>Product result Y11 = "0" | (cond-3)<br>RH_af//RL_tr ≅ RL_tr<br>Product result Y11 = "0" |
| Input voltage V1 = VL<br>Input value X11 = "1" | (cond-2)<br>RL_af//RH_tr ≅ RL_af<br>Product result Y11 = "0" | (cond-4)<br>RH_af//RH_tr ≅ RH_tr<br>Product result Y11 = "1" |

The product result Y**11** shown in Table 3 is a result of product computation of the input value X**11** and the weight value W**11** of the memory cell **100-11**. In one example, the total equivalent resistance value of the memory cell **100-11** may reflect the product result Y**11**.

The output current I**1** of the memory cell **100-11** is equal to the sum of the drain current Id of the transistor tr**11** and the current of the resistor R**11**. In another example, the output current I**1** may also reflect the product result Y**11** of the memory cell **100-11**. When the memory cell **100-11** and other memory cells are connect in series to form a memory string, and the memory string is connected to a load capacitor CL_**1** (not shown in FIG. **3**), the output current of the memory cell **100-11** and the output currents of other memory cells are summed up to form the output current I**1**. The output current I**1** charges the load capacitor CL_**1**, and the charging time T**1** of the load capacitor CL_**1** is related to the output current I**1**. Therefore, the charging time T**1** may also reflect a sum-of-product result. The sum-of-product result of the memory string is a sum of the product result Y**11** of each of the memory cells (which will be described in detail in embodiments in following paragraphs with FIG. **4**).

When the data width of the input value X11 and the weight value W11 is one bit, the product computation of the input value X11 and the weight value W11 is equivalent to a logic "AND" operation.

In the first condition (cond-1) in Table 3, the input voltage V1 is equal to the second input voltage VH, and the total equivalent resistance value of the memory cell 100-11 is substantially equal to the second resistance value RL_af. Therefore, the output current I1 is related to the input voltage V1 divided by the second resistance value RL_af. Since the input voltage V1 is equal to the second input voltage VH which is relatively high, the second resistance value RL_af is a lower resistance value, therefore, the output current I1 has a higher current value, resulting in a shorter charging time T1. In the first condition (cond-1), the input value X11 is "0" and the weight value W11 is "O", hence the product result Y11 is "0", and the product result Y11 is reflected by the shorter charging time T1.

In the second condition (cond-2), the input voltage V1 is equal to the lower first input voltage VL, and the total equivalent resistance value of the memory cell 100-11 is substantially equal to the lower second resistance value RL_af. Therefore, the current value of the output current I1 in the second condition (cond-2) is slightly lower than that in the first condition (cond-1), resulting in a slight increase in the charging time T1. In the second condition (cond-2), the input value X11 is "1" and the weight value W11 is "0", hence the product result Y11 is "O".

In the third condition (cond-3), the input voltage V1 is equal to the higher second input voltage VH, and the total equivalent resistance value of the memory cell 100-11 is substantially equal to the lower second equivalent resistance value RL_tr, therefore, the current value of the output current I1 in the third condition (cond-3) is slightly increased as compared to that of the second condition (cond-2), resulting in a decrease of the charging time T1. In the third condition (cond-3), the input value X11 is "0" and the weight value W11 is "1", hence the product result Y11 is "O".

In the fourth condition (cond-4), the input voltage V1 is equal to the lower first input voltage VL, and the total equivalent resistance value of the memory cell 100-11 is substantially equal to the higher first equivalent resistance value RH_tr, therefore, the current value of the output current I1 in the fourth condition (cond-4) is greatly lower than that in the third condition (cond3), resulting in a greatly increase in the charging time T1. In the fourth condition (cond-4), the input value X11 is "1" and the weight value W11 is "1", hence the product result Y11 is "i", and the product result Y11 is reflected by the greatly increased charging time T1.

FIG. 4 is a schematic diagram of the memory string 10-1 performing computations according to an embodiment of the disclosure. As afore-mentioned, the memory cell 100-11 and other memory cells are connected in series to form a memory string. The source s11 of transistor tr11 of memory cell 100-11 may be coupled to the drain of transistor of neighboring memory cell. Drain current of transistor and current of resistor of each memory cell in the memory string are summed up to form the output current I1. The computation result of sum-of-product computation performed by the memory string may be calculated according to the output current I1. More particularly, as shown in FIG. 4, the memory string 10-1 includes N memory cells 100-11~100-1N. Each of the memory cells 100-11~100-1N is the same as the memory cell 100-11 in FIG. 1. For example, the memory cell 100-12 includes a transistor tr12 and a resistor R12 connected in parallel, the memory cell 100-13 includes a transistor tr13 and a resistor R13 connected in parallel, and so on. The memory string 10-1 is used to perform a sum-of-product computation, which sums up the respective product results Y11~Y1N of the memory cells 100-11~100-1N, so as to obtain a sum-of-product result YS_1.

The memory cells 100-11~100-1N are connected in series. For example, the source s11 of the transistor tr11 of the memory cell 100-11 is connected to the drain d12 of the transistor tr12 of the next neighboring memory cell 100-12, and the second end e2 of the resistor R11 is connected to the first end e3 of the resistor R12. Similarly, the source s12 of the transistor tr12 of the memory cell 100-12 is connected to the drain d13 of the transistor tr13 of the next neighboring memory cell 100-13, and the second end e4 of the resistor R12 is connected to the first end e5 of the resistor R13, and so forth.

The gate g11 of the transistor tr11 is connected to the first signal line WL1 to receive the input voltage V1, and the gate g12 of the next neighboring transistor tr12 is connected to the first signal line WL2 to receive the input voltage V2, and so on, the gate g1N of the Nth transistor tr1N (i.e., the last transistor in the memory string 10-1) is connected to the first signal line WLN to receive the input voltage VN. These first signal lines WL1~WLN are, for example, word lines.

For the first memory cell 100-11, the drain d11 of the transistor tr11 and the first end e1 of the resistor R11 are connected to the second signal line BL1, and the transistor tr11 and the resistor R11 receive the read voltage Vread through the second signal line BL1. The second signal line BL1 is, for example, a bit line.

For the Nth memory cell 100-1N (i.e., the last memory cell in the memory string 10-1), the source s1N of the transistor tr1N and the second end e2N of the resistor R1N are connected to the third signal line SL1. The memory cell 100-1N transmits the output current I1 through the third signal line SL1. The third signal line SL1 is, for example, a source line. The output current I1 and the output voltage Vout1 of the memory string 10-1 are transmitted to a sensing line SS1.

The first end eC1 of the load capacitor CL_1 is connected to the third signal line SL1 to receive the output current I1. That is, the source s1N of the transistor tr1N of the last memory cell 100-1N in the memory string 10-1 and the second end e2N of the resistor R1N are connected to the first end eC1 of the load capacitor CL_1, and the load capacitor CL_1 receives the output current I1 of the memory string 10-1. The output current I1 charges the load capacitor CL_1. The charging time T1 of the load capacitor CL_1 reflects the sum-of-product result YS_1 of the memory string 10-1.

The memory string 10-1 cooperates with the comparator 200. The input end 201 of the comparator 200 is connected to the first end eC1 of the load capacitor CL_1 through the sense line SS1. The input end 201 of the comparator 200 receives the voltage of the first end eC1 of the load capacitor CL_1 (which is equal to an output voltage Vout1). Another input end 202 of the comparator 200 receives a reference voltage Vref, and the comparator 200 compares the output voltage Vout1 with the reference voltage Vref.

At a starting time point t0, the memory string 10-1 starts to operate, and the output current I1 starts to charge the load capacitor CL_1. During the charging process, the voltage at the first end eC1 of the load capacitor CL_1 (which is equal to the output voltage Vout1) continues to increase. Then, at an ending time point t1, the voltage of the first end eC1 of the load capacitor CL_1 reaches the reference voltage Vref (i.e., the output voltage Vout1 is equal to the reference voltage Vref), indicating that the charging-level of the load capacitor CL_1 has met a predefined condition. The time difference between the ending time point t1 and the starting time point t0 may be defined as the charging time T1 of the load capacitor CL_1. The charging time T1 reflects the sum-of-product result YS_1 of memory string **10-1**.

FIG. **5** is a schematic diagram of the memory device **1000** performing computations according to an embodiment of the disclosure. As shown in FIG. **5**, the memory device **1000** includes M memory strings **10-1**~**10-**M, and these memory strings **10-1**~**10-**M form a memory array, with M being "24", for example. Each of the memory strings **10-1**~**10-**M is the same as the memory string **10-1** in FIG. **4**. For example, the memory string **10-2** includes N memory cells **100-21**~**100-2**N. N is "32", for example. The memory strings **10-1**~**10-**M are arranged in parallel. The total number of all memory cells **100-11**~**100-**MN included in the memory device **1000** is M times N, for example, "32" times "24", which is equal to "768".

The memory cell **100-11** of the memory string **10-1**, the memory cell **100-21** of the memory string **10-2**, . . . , and the memory cell **100-**M1 of the memory string **10-**M are jointly connected to the second signal Line BL1 to receive the read voltage Vread.

The memory cell **100-1**N of the memory string **10-1**, the memory cell **100-2**N of the memory string **10-2**, . . . , and the memory cell **100-**MN of the memory string **10-**M are jointly connected to the sensing line SS1, and connected to the first end eC1 of the load capacitor CL_1 through the sensing line SS1. Memory string **10-1** generates output current I1, memory string **10-2** generates output current I2, and so on, memory string **10-**M generates output current IM. The output currents I1~IM are transmitted to the sensing line SS1 and summed to form a total output current IS_1. The total output current IS_1 charges the load capacitor CL_1.

The memory device **1000** is used to perform a sum-of-product computation, and sum up the sum-of-product results YS_1~YS_M of the memory strings **10-1**~**10-**M to obtain a total sum-of-product result YST. The charging time TS1 for the total output current IS_1 to charge the load capacitance CL_1 may reflect the total sum-of-product result YST.

In this embodiment, the component parameters and operating parameters of the memory device **1000** are listed as follows. The effective length of the channel under the oxide layer of respective transistors tr11~trMN of the memory cells **100-11**~**100-**MN is e.g., 0.18 μm, and the effective width is e.g., 0.3 μm. The first resistance value RH_af of respective resistors R11~RMN of the memory cells **100-11**~**100-**MN is e.g., 100MΩ, and the second resistance value RL_af is e.g., 1KΩ. The read voltage Vread received by the memory device **1000** through the second signal line BL1 is, for example, 0.5V. The capacitance value of the load capacitor CL_1 is e.g., 1 pf. The reference voltage Vref of the comparator **200** is e.g., 0.3V. When the voltage of the first end eC1 of the load capacitor CL_1 reaches 0.3V, the ending time point t1 is recorded, and the charging time T1 of the load capacitor CL_1 is computed accordingly.

FIG. **6** is a measurement diagram of the computation results of the memory device **1000** in FIG. **5**. The value on the horizontal axis of the measurement diagram in FIG. **6** represents the total sum-of-product result YST of sum-of-product computation performed by the memory device **1000**, and the value on the vertical axis represents the charging time T1 of the load capacitor CL_1.

The measurement diagram includes a plurality of measurement points, and these measurement points are substantially divided into two types, including: a plurality of first measurement points m1 of a first type and a plurality of second measurement points m2 of a second type. Each of the first measurement points m1 is shown as a "thick black dot" and each of the second measurement points m2 is shown as a "short cross lines". When the weight values W11~WMN of the memory cells **100-11**~**100-**MN of the memory device **1000** are all "1", the computation results of the memory device **1000** are displayed as the first measurement points m1. For example, one first measurement point m1-1 indicates that: the total sum-of-product result YST of the memory device **1000** is equal to "700", and the charging time T1 of the load capacitor CL_1 is equal to 0.8 μs.

On the other hand, when half of the weight values W11~WMN of the memory cells **100-11**~**100-**MN of the memory device **1000** are "1", the computation results of the memory device **1000** are displayed as the second measurement points m2.

The first measurement points m1 and the second measurement points m2 substantially show a linear distribution, indicating that the charging time T1 of the load capacitor CL_1 is related to the total sum-of-product result YST of the memory device **1000**.

It will be apparent to those skilled in the art that various modifications and variations may be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

What is claimed is:

**1**. A memory device, for performing an in-memory computation, the memory device comprising:

a plurality of memory cells, forming a plurality of memory strings each performing a sum-of-product computation that is a sum operation for a product computation of each of the memory cells, each of the memory cells stores a weight value, and each of the memory cells comprising:

a transistor, having a gate, a drain and a source, the gate receives an input voltage, the input voltage indicates an input value, when the transistor operates at a first operating point, the input voltage is equal to a first input voltage, when the transistor operates at a second operating point, the input voltage is equal to a second input voltage, the second input voltage is higher than the first input voltage; and

a resistor, connected to the drain and the source of the transistor, when the resistor operates in a first state, the weight value is equal to a first weight value, when the resistor operates in a second state, the weight value is equal to a second weight value,

wherein, each of the memory cells performs the product computation of the input value and the weight value, and each of the memory strings is respectively connected to a load capacitor and generates an output current, and a charging time for the output current to charge the load capacitor reflects a sum-of-product result of the sum-of-product computation.

**2**. The memory device according to claim **1**, wherein, the first weight value is a bit of "1", the second weight value is a bit of "0", when the input voltage is equal to the first input voltage, the input value is a bit of "1", when the input voltage is equal to the second input voltage, the input value is a bit of "0".

**3**. The memory device according to claim **2**, wherein, the first input voltage is higher than a product of a threshold voltage of the transistor and a first ratio.

4. The memory device according to claim 1, wherein, the resistor is an anti-fuse, the first state is an initial state, and the second state is a programmed state, when the resistor operates in the initial state, the resistor has a first resistance value, when the resistor operates in the programmed state, the resistor has a second resistance value, and the first resistance value is higher than the second resistance value.

5. The memory device according to claim 4, wherein, when the transistor operates at the first operating point, the transistor has a first equivalent resistance value, when the transistor operates at the second operating point, the transistor has a second equivalent resistance value, the first equivalent resistance value is higher than the second equivalent resistance value.

6. The memory device according to claim 5, wherein, the first equivalent resistance value is lower than or equal to a product of the second equivalent resistance value and a second ratio.

7. The memory device according to claim 5, wherein:

the first resistance value is higher than the first equivalent resistance value;

the first equivalent resistance value is higher than the second equivalent resistance value; and

the second equivalent resistance value is higher than the second resistance value.

8. The memory device according to claim 5, wherein:

the first resistance value is higher than the first equivalent resistance value;

the first equivalent resistance value is higher than the second resistance value; and

the second resistance value is higher than the second equivalent resistance value.

9. An operating method of a memory device, for performing an in-memory computation, the memory device comprises a plurality of memory cells which form a plurality of memory strings each performing a sum-of-product computation that is a sum operation for a product computation of each of the memory cells, each of the memory cells comprises a transistor and a resistor, the transistor has a gate, a drain and a source, the resistor is connected to the drain and the source of the transistor, and the operating method comprising:

receiving an input voltage through the gate of the transistor, the input voltage indicates an input value;

setting the input voltage as a first input voltage, so as to operate the transistor at a first operating point;

setting the input voltage as a second input voltage, so as to operate the transistor at a second operating point, the second input voltage is higher than the first input voltage;

storing a weight value by each of the memory cells;

operating the resistor in a first state, so as to set the weight value as a first weight value;

operating the resistor in a second state, so as to set the weight value as a second weight value; and

performing the product computation of the input value and the weight value by each of the memory cells,

wherein each of the memory strings is respectively connected to a load capacitor and generates an output current, and a charging time for the output current to charge the load capacitor reflects a sum-of-product result of the sum-of-product computation.

10. The operating method according to claim 9, wherein, the first weight value is a bit of "1", the second weight value is a bit of "0", when the input voltage is equal to the first input voltage, the input value is a bit of "1", when the input voltage is equal to the second input voltage, the input value is a bit of "0".

11. The operating method according to claim 10, wherein, the first input voltage is higher than a product of a threshold voltage of the transistor and a first ratio.

12. The operating method according to claim 9, wherein, the resistor is an anti-fuse, the first state is an initial state, and the second state is a programmed state, when the resistor operates in the initial state, the resistor has a first resistance value, when the resistor operates in the programmed state, the resistor has a second resistance value, and the first resistance value is higher than the second resistance value.

13. The operating method according to claim 12, wherein, when the transistor operates at the first operating point, the transistor has a first equivalent resistance value, when the transistor operates at the second operating point, the transistor has a second equivalent resistance value, the first equivalent resistance value is higher than the second equivalent resistance value.

14. The operating method according to claim 13, wherein, the first equivalent resistance value is lower than or equal to a product of the second equivalent resistance value and a second ratio.

15. The operating method according to claim 13, wherein:

the first resistance value is higher than the first equivalent resistance value;

the first equivalent resistance value is higher than the second equivalent resistance value; and

the second equivalent resistance value is higher than the second resistance value.

16. The operating method according to claim 13, wherein:

the first resistance value is higher than the first equivalent resistance value;

the first equivalent resistance value is higher than the second resistance value; and

the second resistance value is higher than the second equivalent resistance value.

* * * * *